LEVEL 1 — GENERAL NORMALIZED EXPRESSION

$$\mathrm{MP}_{\mathrm{norm}} = [(\mathrm{MI} \times \mathrm{NP}) \times \mathrm{SW} \times \mathrm{MS}^{\gamma} \times \mathrm{CP}^{\alpha}]^{1/n_{\mathrm{eff}}}$$

$n_{\mathrm{eff}} = 3 + \gamma + \alpha$ $\quad \mathrm{MP}_{\mathrm{norm}} \in (0, 1]$ $\quad$ all components $\in (0, 1]$

Weighted geometric mean with weights $\omega_{\mathrm{MI}} = \omega_{\mathrm{NP}} = \omega_{\mathrm{SW}} = 1/n_{\mathrm{eff}}$, $\omega_{\mathrm{MS}} = \gamma/n_{\mathrm{eff}}$, $\omega_{\mathrm{CP}} = \alpha/n_{\mathrm{eff}}$

γ: weight of the security environment $\quad$ α: weight of cyber dependence, not a measure of capability

LEVEL 2 — COMPONENT DISAGGREGATION

MI

Maritime Interests

What the state seeks to obtain, protect, or project through the sea.

$$\mathrm{MI} = \Sigma_i \, w_i \cdot \mathrm{MI}_i$$

$\Sigma \, w_i = 1$; $\mathrm{MI} \in (0, 1]$

Weighted linear sub-index: interests are substitutable and w_i act as rates of substitution.

Colombia: n = 18 maritime interests.

NP

Naval Power

Capacity to act on the maritime domain to defend interests, deter threats, and project influence.

$$\mathrm{NP} = F \times P \times \mathrm{MDA}^{\mu}$$

$\mathrm{NP} \in (0, 1]$; μ: MDA calibration

F: force, including tactical-operational maritime cyberdefense.

P: position. MDA: coverage and quality of the maritime picture.

SW

Strategic Will

Capacity to turn maritime interests into sustained decisions, capabilities, and actions.

$$\mathrm{SW} = (\mathrm{PW}^{w_1} \cdot \mathrm{IW}^{w_2} \cdot \mathrm{SocW}^{w_3})^{1/\Sigma w}$$

$\mathrm{SW} \in (0, 1]$; multiplier of the whole system

PW: political, IW: institutional, SocW: social will.

Maritime consciousness, a cultural construct distinct from MDA, enters through SocW.

MS

Maritime Security

Condition of the strategic environment: it measures the incidence of threats, not capabilities.

$$\mathrm{MS} = (\mathrm{IC}^{a_1} \cdot \mathrm{TE}^{a_2} \cdot \mathrm{BC}^{a_3})^{1/\Sigma a}$$

$\mathrm{MS} \in (0, 1]$; global weight γ

IC: interstate conflicts and gray zone. TE: terrorism and extremism. BC: blue crime.

Cyber vectors of each threat count here.

CP

Cyber Power

Capacity to protect, govern, and restore the cyber assets of the maritime domain, amplified by diplomatic projection.

$$\mathrm{CP} = \min\{1, (\beta_1 \, \mathrm{CS_m} + \beta_2 \, \mathrm{CD_m} + \beta_3 \, \mathrm{RC_m})(1 + \delta \cdot \mathrm{CDip})\}$$

$\mathrm{CP} \in (0, 1]$; $\Sigma \, \beta_i = 1$; global weight α

$\mathrm{CS_m}$: maritime cybersecurity. $\mathrm{CD_m}$: strategic maritime cyberdefense.

$\mathrm{RC_m}$: maritime recovery capacity, $\min\{1, t_{\mathrm{ref}}/t\}$. CDip: cyber diplomacy, amplifier only.

OPERATIONAL BOUNDARIES BETWEEN COMPONENTS

- **Cyberdefense:** execution of operations (active defense, military cyber intelligence, offensive capabilities) in F; governance, cooperation, doctrine, and deterrent posture in $\mathrm{CD_m}$.
- **Protection:** controls and information sharing over maritime IT/OT assets in $\mathrm{CS_m}$; restoration after an incident in $\mathrm{RC_m}$.
- **Awareness:** MDA measures the maritime picture; the protection of the systems that produce it is measured in $\mathrm{CS_m}$.
- **Cyber diplomacy:** enters only through CDip; excluded from PW and from defense cooperation in $\mathrm{CD_m}$.
- **Threats:** their incidence, including cyber vectors, only in MS; capabilities against them only in NP and CP.